01 / PROBLEM AND CONTRIBUTION

Gauss-Seidel load-flow study

Power System Analysis · FISAC case study | Undergraduate | Two-person study

RESULT AT A GLANCE

Documented an iterative load-flow procedure from flat-start initialization to convergence checks and line-power calculations, distinguishing slack, PV, and PQ bus specifications.

The engineering question

How can an iterative method estimate steady-state bus voltages and power flow?

Situation

Load-flow analysis estimates steady-state bus voltages and line power from specified network quantities. This case study develops the Gauss-Seidel process and its convergence considerations.

Task

Co-author the bus-type explanation, voltage-update procedure, stopping condition, and discussion of the method's limitations.

My contribution

Co-authored case study with B. K. V. Surya.

Tools and concepts

Power systems / Load-flow analysis / Gauss-Seidel / Numerical methods / Technical analysis

02 / DESIGN AND METHOD

From the task to implementation

01 / Implementation step

Distinguished slack, PV, and PQ buses by their specified and unknown quantities. Developed the Gauss-Seidel procedure for the case where non-slack buses are PQ buses.

02 / Implementation step

Outlined flat-start voltage initialization, successive voltage updates, a tolerance-based stopping condition, and subsequent line-power calculations.

Specified and unknown quantities

The report distinguishes slack, PV, and PQ buses and then outlines the case in which non-slack buses are PQ buses. A flat-start voltage estimate initializes successive updates toward a tolerance-based stopping condition.

Architecture / workflow

Conceptual summary of the documented workflow; not a simulation result.

01	Bus specifications
02	Y-bus matrix
03	Voltage iteration
04	Power-flow evaluation

03 / RESULTS AND ENGINEERING PERSPECTIVE

Evidence, interpretation, and next steps

Reported result

The supplied report explains the algorithm and compares its simplicity and memory requirements with its convergence limitations.

Convergence is part of the method

An iterative procedure needs a stopping rule and a record of its convergence behavior. The supplied work explains the algorithm and its tradeoffs; it does not establish a benchmark network result or execution-time improvement.

Scope and limitations

The study emphasizes algorithm structure and convergence tradeoffs. Its Gauss-Seidel approach is most useful as a small-system learning example; a network benchmark would be a separate extension.

Possible next steps - proposed, not completed results

- Solve a fully specified small network and report residuals, iterations, and the stopping tolerance.
- Compare against an independently checked reference solution using the same network data.

Source material: POWER SYSTEM ANALYSIS FISAC.pdf in gptz/PSA.

Prepared from the supplied reports, project files, and verified portfolio copy. Figures retain their original content; explanatory diagrams are labelled.